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Jashore-7408, Bangladesh

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Campus Based Donation Management System

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This project titled **Campus Based Donation Management System** submitted by **Ramjan Ali Kha** for progress defence for the degree of B.Sc. in Computer Science and Engineering on November 23, 2025.

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List of Acronyms

Table 1: **List of Acronyms.**

Acronyms	**Full Form**
DMS	Donation Management System
UI	User Interface
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
JWT	JSON Web Token
ORM	Object-Relational Mapping
REST	Representational State Transfer
HTTPS	Hypertext Transfer Protocol Secure
SQL	Structured Query Language
EF	Entity Framework
RBAC	Role Based Access Control
HTML	Hypertext Markup Language
CSS	Cascading Style Sheets
TypeScript	Programming Language
React	JavaScript Library
Redux	State Management
.NET	Framework
C#	Programming Language

Abstract

Non-governmental organizations and charitable institutions must execute their operational and financial activities efficiently to achieve their objectives. These methods encompass campaign coordination, transparent fundraising and resource distribution, and efficient volunteer administration. Nonetheless, numerous companies continue to depend on paper-based procedures and manual recordkeeping. While participating in fundraising initiatives for the 2024 Sylhet flood and the medical bills of our late sister Atika (CSE 17-18), I noted that these campaigns frequently result in inaccuracies, insufficient transparency, and superfluous bureaucratic complexities. The proposed Donation Management System (DMS) consolidates all pertinent functions into a unified, user-friendly software platform to tackle these difficulties.

The system employs a novel, multi-layered architecture. ASP.NET Core Web API is utilized to establish a safe and resilient infrastructure, whereas React and TypeScript are employed to create a responsive and user-centric frontend. The solution employs Entity Framework Core alongside a relational database to provide effective data management and scalability.

Security is essential and includes technologies such as BCrypt password hashing, CSRF protection, input validation, and secure payment processing. The responsive design guarantees interoperability across all devices, encompassing personal computers and mobile phones. The Donation Management System markedly decreases manual labor, improves transparency, and streamlines the implementation of donation campaigns for diverse organizations.

The project includes features such as campaign management, donation processing, volunteer coordination, user authentication and access restriction, feedback sentiment analysis, and a physical donation tracking system. This technology aims to increase donor confidence and participation by promoting transparency and automating routine tasks to simplify the donation process.

Chapter 1

Introduction

Chapter 1: Introduction

1.1 Overview

The **Campus Based Donation Management System** is a comprehensive, web-based software platform developed to digitize, optimize, and automate the operational workflows associated with donation collection, campaign administration, volunteer coordination, and donor relationship management within an institutional context. The system establishes a unified digital environment that facilitates structured interaction among three primary user roles — Administrators, Donors, and Volunteers — through an intuitive and role-adaptive interface, thereby eliminating the inefficiencies inherent in fragmented or manual approaches.

The platform delivers a broad set of core functionalities, encompassing real-time donation tracking, campaign lifecycle management, integrated online payment processing via SSLCommerz, physical donation verification through OTP-based confirmation, volunteer activity reporting with a structured rank progression framework, expense voucher submission and approval workflows, and a controlled fund withdrawal and reserve management mechanism. Beyond these operational capabilities, the system incorporates a machine learning subsystem that provides donation trend forecasting, anomaly detection, donor attrition prediction, and campaign success likelihood assessment — enabling data-driven strategic decision-making for administrators.

From an architectural standpoint, the system is built upon a modern, multi-tier technology stack: React with TypeScript constitutes the responsive and component-based frontend; ASP.NET Core Web API provides the scalable and RESTful backend infrastructure; and Microsoft SQL Server, accessed through Entity Framework Core, serves as the relational data persistence layer. Security is enforced at multiple levels through JSON Web Token (JWT) authentication, BCrypt password hashing, Cross-Site Request Forgery (CSRF) protection, server-side input validation, and role-based access control middleware. Automated communication is facilitated via integrated email and SMS notification services, ensuring timely delivery of transactional alerts and system updates to all stakeholders.

The system's modular and layered architecture ensures long-term scalability, maintainability, and extensibility, providing a robust foundation for future enhancements and integrations.

1.2 Background

Charitable organizations and donation management initiatives have always relied on manual, paper-based systems or fragmented digital solutions to monitor donations, oversee campaigns, and coordinate volunteers. These traditional methods are frequently laborious, prone to errors, and ineffective. Common difficulties intrinsic to these systems include:

- **Tracking Latency:** Challenges in monitoring donations and campaign advancement in real-time.
- **Data Redundancy:** Manual data entry procedures resulting in errors and inconsistencies in data.
- **Operational Inefficiency:** Difficulties in coordinating volunteer efforts and accurately monitoring their contributions.
- **Analytical Deficiencies:** A constrained capacity to assess donor behavior and appraise campaign efficacy.
- **Obstacles to Communication:** An absence of clear communication routes between donors and organizations.
- **Lack of Transparency:** Donors are often unaware of how their contributions are utilized, reducing trust and long-term engagement with organizations.
- **No Financial Accountability:** The absence of audit trails and financial reporting mechanisms makes it difficult to verify that funds are allocated as intended.
- **Limited Donor Visibility:** Donors cannot access real-time information on campaign progress, fund utilization, or organizational impact, discouraging repeat contributions.

- **Absence of Physical Donation Management:** Conventional systems lack structured mechanisms for logging, verifying, and tracking in-kind or physical donations, resulting in unrecorded contributions and diminished organizational accountability.

- **Inadequate Expense Voucher Oversight:** The absence of a formalized voucher submission and approval workflow impedes the ability to monitor how campaign-related expenditures are incurred and authorized by volunteers or staff.

- **Unregulated Fund Withdrawal Processes:** Without a controlled withdrawal and reserve fund management mechanism, organizations are susceptible to misappropriation of funds and lack a verifiable audit trail for financial disbursements.

- **Absence of Predictive and Analytical Capabilities:** Existing systems are predominantly reactive, offering no means to forecast future donation trends, identify anomalous transactions, assess the likelihood of campaign success, or predict donor attrition — capabilities that are increasingly critical for strategic organizational planning.

- **Lack of Volunteer Recognition and Incentivization:** The absence of structured rank progression and achievement acknowledgement frameworks diminishes volunteer motivation, reducing long-term engagement and retention within donation-driven initiatives.

- **Insufficient Automated Notification Infrastructure:** Manual communication workflows fail to deliver timely and reliable notifications — such as donation confirmations, one-time password delivery via SMS, and campaign status updates via email — thereby degrading the overall responsiveness and trustworthiness of the system.

Due to the swift integration of digital technology and escalating stakeholder demands for online services, companies acknowledge the necessity for a centralized, automated platform. The Donation Management System fills this void by offering a thorough digital solution that automates critical procedures while ensuring confidentiality, transparency, and user accessibility.

1.3 Motivation

The motivation behind developing the Donation Management System stems from the evident inefficiencies in current donation management processes. Many organizations struggle to:

- Manage multiple donation channels and payment methods simultaneously.
- Track volunteer activities and measure their impact accurately.
- Analyze campaign performance and donor satisfaction through feedback.
- Provide donors with real-time updates on campaign progress.
- Maintain secure records of financial transactions and donor information.

The project aims to create a modern, scalable solution that not only addresses these challenges but also provides a foundation for future innovations in donation management. By leveraging contemporary web technologies and implementing best practices in software architecture, the system ensures reliability, security, and user satisfaction.

1.4 Problem Statement

Organizations managing donation campaigns face numerous challenges with manual and fragmented systems:

- **Lack of Real-Time Tracking:** Donors and organizations cannot track campaign progress and donation status in real-time.
- **Manual Payment Processing:** Collecting donations through manual methods is inefficient, error-prone, and insecure.
- **Volunteer Coordination:** Lack of a centralized system makes it difficult to coordinate volunteers, track their activities, and measure their contributions.

- **Data Integrity Issues:** Manual record-keeping leads to data inconsistencies and difficulty in generating accurate reports.
- **Limited Analytics:** Organizations lack insights into donor behavior, campaign effectiveness, and testimonial sentiment.
- **Security Concerns:** Manual systems and unencrypted digital records pose significant risks to sensitive donor and financial data.
- **Lack of Transparency:** Existing systems don't include transparency. Donors feel motivated if they see the outcome and exact amount being expended.
- **No Bookkeeping:** Existing local systems aren't integrated with bookkeeping feature.

An automated, web-based management system is essential to address these issues and provide a secure, transparent, and efficient platform for managing donations and campaigns.

1.5 Questions Addressed by the Project

The project seeks to address the following research questions:

- How can a unified web-based system improve donation collection, tracking, and campaign management?
- What are the benefits of implementing role-based access control for multiple user types in a donation management context?
- How effective is real-time donation tracking in improving campaign transparency and donor satisfaction?
- To what extent does an integrated and secure online payment module streamline the donation process?

- How can sentiment analysis on donor testimonials provide insights into campaign impact and donor satisfaction?
- What database management and security techniques are most effective in protecting sensitive donor information?

1.6 Objectives

The following are the main objectives of the Donation Management System:

- **To provide a unified web-based platform** that streamlines donation collection, campaign management, volunteer coordination, and reporting.
- **To implement role-based access control** with distinct functions for Admins, Donors, Volunteers, and Campaign Creators.
- **To offer real-time donation and campaign tracking** that provides transparency and immediate feedback to stakeholders.
- **To integrate secure payment processing** using SSLCommerz gateway for safe and efficient transaction handling.
- **To improve user experience** with a responsive, intuitive interface accessible across all devices.
- **To ensure data security and integrity** through proper authentication, encryption, and validation mechanisms.
- **To enable sentiment analysis** on donor testimonials to measure campaign impact and donor satisfaction.
- **To provide comprehensive reporting and analytics** for decision-making and performance tracking.

1.7 Organization

The Donation Management System development process is presented methodically in each of the six chapters that make up this project report. The report's structure is as follows:

Chapter 1: Introduction - Overview, background, motivation, problem statement, objectives, and organization.

Chapter 2: Background and Related Technologies - Existing systems, relevant technologies, and identified gaps.

Chapter 3: Methodology - System design, development process, testing, deployment, and security measures.

Chapter 4: System Architecture and Design - Frontend, backend, database design, and system diagrams (ER, Use Case, Schema).

Chapter 5: Implementation - System setup, technologies used, and implementation of key modules.

Chapter 6: Conclusion and Future Work - Summary and recommendations for future enhancements.

Chapter 2

Background and Related Technologies

Chapter 2: Background and Related Technologies

2.1 Overview

As digital transformation accelerates, organizations increasingly adopt automated solutions for improved management and service delivery. Donation management systems have evolved from manual record-keeping to sophisticated digital platforms that enhance operational efficiency and stakeholder engagement [**?**]. This chapter examines existing donation management approaches, contemporary technologies, and identifies gaps that the Donation Management System addresses.

2.2 Traditional Donation Management Methods

Conventional donation management typically relies on manual processes including paper-based records, in-person collection, and spreadsheet tracking. These methods present significant challenges [**?**]:

- Difficulty in real-time donation tracking and campaign monitoring.
- Increased likelihood of errors and data loss.
- Inefficient resource allocation for volunteer coordination.
- Limited ability to analyze campaign effectiveness.
- Security risks associated with physical records and unencrypted digital storage.

2.3 Existing Digital Donation Management Solutions

Several commercial donation management platforms exist, including:

Donorbox: Cloud-based donation platform with payment processing and donor management [**?**].

GiveWP: WordPress-integrated donation plugin with customizable forms.

Network for Good: Comprehensive fundraising and donor management platform [?].

These solutions, while effective for many organizations, often lack customization for specific institutional needs, are costly, and may not integrate seamlessly with existing systems used by smaller organizations [?]. Advanced solutions also explore secure and transparent e-donation systems using technologies like Blockchain [?].

2.4 Role-Based Access Control in Management Systems

Role-Based Access Control (RBAC) is essential for multi-user platforms [?]. RBAC provides:

- **Security:** Restricts access to sensitive information based on user roles.
- **Usability:** Displays only features relevant to each user's responsibilities.
- **System Integrity:** Prevents unauthorized modifications.

The Donation Management System implements RBAC for Admins (full system access), Donors (donation and profile management), Volunteers (activity reporting), and Campaign Creators (campaign management).

2.5 Payment Gateway Integration

Secure online payment processing is critical for donation platforms. SSLCommerz, a popular payment gateway in South Asia, provides:

- Multiple payment method support (credit card, mobile banking, wallet).
- Secure transaction processing and encryption.
- Real-time payment confirmation and callback notifications.
- Comprehensive transaction reporting.

2.6 Sentiment Analysis in Donation Systems

Sentiment analysis provides organizations with insights into donor satisfaction and campaign impact [**?**]. Using TextBlob or similar natural language processing libraries, testimonials can be automatically classified as positive, negative, or neutral [**?**], enabling:

- Measurement of campaign effectiveness through donor feedback.
- Identification of areas for improvement.
- Data-driven decision-making for future campaigns.

2.7 Modern Web Technologies

The Donation Management System leverages contemporary web development technologies [**?**]:

- **Frontend:** React with TypeScript [**?**], Redux Toolkit for state management, Tailwind CSS for styling.
- **Backend:** ASP.NET Core Web API for building scalable RESTful services [**?**].
- **Database:** SQL Server for robust relational data management and Entity Framework Core for ORM [**?**].
- **Authentication:** JWT tokens for stateless authentication and OTP-based email and SMS verification [**?**].
- **Security:** BCrypt for password hashing [**?**], CSRF tokens for form protection, input validation and sanitization.

These technologies are critical for ensuring system scalability and maintainability [**?**].

2.8 Identified Gaps

While existing solutions provide valuable features, gaps remain in:

- Customization for organization-specific workflows and requirements, particularly in campus and higher-education contexts [**?**].
- Integration of machine learning capabilities such as donor attrition prediction and donation forecasting [**?**, **?**].
- Anomaly detection in financial transactions to safeguard campaign integrity [**?**].
- Structured volunteer recognition frameworks to improve retention [**?**].
- Cost-effectiveness for smaller organizations and educational institutions.
- Flexibility in feature selection and deployment options.

The Donation Management System addresses these gaps by providing a customizable, feature-rich platform tailored to diverse organizational needs.

Chapter 3

Methodology

Chapter 3: Methodology

3.1 Overview

This chapter describes the design, development, testing, and implementation of the Donation Management System. It outlines the technological stack, system architecture, and development methodologies employed.

3.2 System Design

Developing a solid, scalable, and intuitive architecture that supports various user roles while preserving security and performance is the major goal of the system design process. The front end, back-end, and database design are the three primary components of the design phase.

3.2.1 Frontend Design

The frontend is built using React with TypeScript, ensuring type safety and code maintainability. Key design principles include:

- **Responsive Design:** Utilizes CSS Flexbox and media queries for cross-device compatibility.
- **Component-Based Architecture:** Reusable React components for efficient development.
- **State Management:** Redux Toolkit for predictable and efficient state management.
- **Accessibility:** Follows WCAG guidelines for inclusive design.

3.2.2 Backend Design

The backend utilizes .NET Core Web API with an MVC architecture:

- **Controllers:** Handle HTTP requests and business logic.

- **Models:** Define data structures and relationships.
- **Services:** Encapsulate business logic and operations.
- **Middleware:** Implement authentication, authorization, and error handling.

3.2.3 Database Design

SQL Server with Entity Framework Core provides:

- Relational data modeling with enforced referential integrity.
- Seven main entities: Users, Campaigns, Donations, Testimonials, Volunteers, VolunteerReports, CampaignUpdates.
- Indexed queries for performance optimization.

3.3 Development Process

The project follows Agile methodology with iterative development sprints, allowing continuous feedback incorporation and incremental feature delivery [?].

3.3.1 Frontend Development

Role-specific interfaces were developed for:

- **Admin:** Dashboard, user management, analytics, campaign management.
- **Donor:** Donation interface, testimonial submission, profile management.
- **Volunteer:** Activity reporting, campaign tracking, hour submission.

3.3.2 Backend Development

Key modules include:

- **Authentication:** JWT-based authentication with role-based authorization.
- **Donation Processing:** Donation creation, SSLCommerz payment integration, transaction tracking.
- **Campaign Management:** CRUD operations, progress tracking, volunteer coordination.
- **Sentiment Analysis:** Testimonial processing with TextBlob for sentiment classification.
- **Analytics:** Aggregation of donation data, campaign metrics, volunteer statistics.

3.4 Testing

Comprehensive testing includes:

- **Unit Testing:** Individual component and function testing.
- **Integration Testing:** Module interaction and API endpoint testing.
- **Security Testing:** Authentication, authorization, and data protection validation.
- **Performance Testing:** Load testing and response time optimization.

3.5 Deployment

Deployment considerations include:

- Frontend deployment to CDN or static hosting.
- Backend deployment to cloud services (Azure, AWS).
- Database configuration and migration management.

- SSL certificate installation for HTTPS security.
- Environment configuration and secrets management.

3.6 Security Measures

Security is implemented throughout the system:

- **Authentication:** JWT tokens with secure token refresh mechanisms.
- **Password Security:** BCrypt hashing with salt.
- **CSRF Protection:** Token-based CSRF prevention.
- **Input Validation:** Server-side and client-side validation.
- **Data Encryption:** HTTPS for data transmission, encrypted storage for sensitive data.
- **Access Control:** Role-based authorization middleware.

Chapter 4

System Architecture and Design

Chapter 4: System Architecture and Design

4.1 Overview

This chapter describes the Donation Management System's architecture, encompassing frontend design, backend architecture, database structure, and system diagrams that illustrate system components and their interactions.

4.2 Frontend Architecture

The frontend employs a component-based architecture with Redux for state management:

- React components for UI rendering.
- Redux store for centralized state management.
- Tailwind CSS for styling and responsiveness.
- TypeScript for type safety and code quality.

4.3 Backend Architecture

The .NET Core Web API follows RESTful principles with clear separation of concerns:

- Controllers handle HTTP requests and responses.
- Services implement business logic and operations.
- Repositories abstract data access operations.
- Middleware handles cross-cutting concerns.

4.4 Database Schema

The system uses seven primary entities:

- **Users:** User accounts with role assignments and authentication data.
- **Campaigns:** Fundraising campaigns with targets and progress tracking.
- **Donations:** Individual donation records linked to campaigns and donors.
- **Testimonials:** Donor testimonials with ratings and sentiment scores.
- **Volunteers:** Volunteer profiles and skill information.
- **VolunteerReports:** Activity logs and hour submissions.
- **CampaignUpdates:** Progress updates and campaign communications.

4.5 ER Diagram

The Entity-Relationship diagram illustrates the relationships between system entities and their attributes, showing one-to-many and one-to-one relationships between tables.

images/er_diagram.png

Figure 4.1: **Entity-Relationship Diagram.**

4.6 Use Case Diagram

The Use Case diagram depicts interactions between user roles (Admin, Donor, Volunteer, Campaign Creator) and system functionalities:

images/use_case_diagram.png

Figure 4.2: **Use Case Diagram.**

4.7 Schema Diagram

The Database Schema diagram shows the detailed structure of all tables, including fields, data types, and relationships:

images/schema_diagram.png

Figure 4.3: **Database Schema Diagram.**

4.8 Authentication Flow

The system implements JWT-based authentication:

- User credentials verified against hashed passwords.
- JWT token issued upon successful authentication.

- Token included in subsequent requests for authorization.
- Role-based middleware validates user permissions.

4.9 Payment Processing Flow

Integration with SSLCommerz gateway:

- Donation request initiated by user.
- Redirect to SSLCommerz payment gateway.
- Payment processed securely.
- Callback notification received and verified.
- Donation record created upon successful payment.

Chapter 5

System Implementation

Chapter 5: System Implementation

5.1 System Setup

The implementation of the Donation Management System involves deploying a comprehensive web application with distinct frontend and backend components.

5.1.1 Technologies Used

Frontend Technologies:

Table 5.1: **Frontend Technologies.**

SN	Technology
1	React
2	TypeScript
3	Redux Toolkit
4	Tailwind CSS
5	Axios (HTTP Client)
6	Vite (Build Tool)

Backend Technologies:

Table 5.2: **Backend Technologies.**

SN	Technology
1	.NET Core 7/8
2	C#
3	Entity Framework Core
4	SQL Server
5	ASP.NET Core Identity
6	JWT Authentication

5.2 System Implementation

5.2.1 Dashboard Overview

The Admin Dashboard provides a comprehensive overview of system metrics, campaign status, donation progress, and volunteer activity. It displays real-time statistics for informed decision-making.

images/admin_dashboard.png

Figure 5.1: **Admin Dashboard.**

5.2.2 Campaign Management

Campaign creators can establish new campaigns, set fundraising targets, upload promotional materials, track progress, and post updates. The system monitors campaign status and provides real-time fundraising metrics.

5.2.3 Donation Processing

The donation module integrates SSLCommerz payment gateway, supporting multiple payment methods. Donors can make contributions securely, receive confirmations, and track their donation history through their profile.

5.2.4 Volunteer Management

Volunteers can register, submit activity reports with hour tracking, and monitor their contributions. Admins can approve submissions, track volunteer impact, and generate volunteer activity reports.

5.2.5 Testimonial and Sentiment Analysis

The system collects donor testimonials with ratings and feedback. Sentiment analysis automatically classifies testimonials, providing organizations with insights into donor satisfaction and campaign effectiveness.

5.2.6 User Authentication

Secure user authentication implements JWT tokens with email verification. Users authenticate with credentials, receive JWT tokens, and maintain sessions securely with automatic token refresh.

5.2.7 Analytics and Reporting

The Analytics module provides comprehensive reports on donation trends, campaign performance, volunteer contributions, and donor demographics. Data visualization helps stakeholders understand system performance and impact.

images/analytics_dashboard.png

Figure 5.2: **Analytics Dashboard.**

5.3 Key Features Implementation

5.3.1 Donation Tracking

Real-time donation tracking displays campaign progress toward targets, tracks individual donations, and provides transparency to donors and campaign creators.

5.3.2 Campaign Updates

Campaign creators can post updates about campaign progress, impact stories, and milestone achievements to keep donors informed and engaged.

5.3.3 Volunteer Reporting

Volunteers submit detailed activity reports including hours contributed, descriptions of work performed, and associated campaigns, enabling proper volunteer management and recognition.

5.3.4 Payment Integration

SSLCommerz integration enables secure payment processing with support for multiple payment methods, automatic transaction confirmation, and secure callback handling.

5.3.5 Email Notifications

Automated email notifications inform users of donation confirmations, campaign updates, volunteer report approvals, and account notifications.

Chapter 6

Conclusion and Future Work

Chapter 6: Conclusion and Future Work

6.1 Conclusion

The Donation Management System represents a comprehensive digital solution for modern donation and campaign management. By automating critical operations, providing real-time transparency, and implementing advanced features like sentiment analysis, the system significantly enhances organizational efficiency and stakeholder engagement.

The platform successfully addresses traditional challenges in donation management through:

- Centralized donation processing with secure payment integration.
- Real-time campaign tracking and progress monitoring.
- Efficient volunteer coordination and activity management.
- Data-driven insights through analytics and sentiment analysis.
- Role-based access control ensuring security and appropriate feature access.
- Responsive design for accessibility across all devices.

Built on modern, scalable technologies (.NET Core, React, SQL Server), the system provides a robust foundation for current operations while remaining flexible for future enhancements. The implementation of industry-standard security practices ensures donor data protection and system integrity, fostering trust among users and stakeholders.

6.2 Future Work

The existing system provides a solid foundation for donation management. Future enhancements could include:

- **Bookkeeping Integration:** Integration with accounting and bookkeeping systems for automated financial record-keeping, expense tracking, and tax reporting.

- **Advanced Sentiment Analysis:** Enhanced sentiment analysis capabilities including multilingual support, real-time feedback analysis, and predictive donor satisfaction metrics.

These enhancements will strengthen the system's financial management capabilities and provide deeper insights into donor engagement and satisfaction.

References